Closed Loop Control of Single-phase Boost PFC Converter-IV

Introduction

In the realm of power electronics, ensuring efficient and stable power delivery is paramount. Single-phase Boost Power Factor Correction (PFC) converters play a critical role in achieving this by shaping the input current to be in phase with the input voltage, thereby achieving unity power factor. This lecture delves into the intricate details of designing the closed-loop control for such converters, building upon previous discussions of their fundamental operation. The primary objectives of this control system are twofold: first, to precisely regulate the output voltage to its desired reference value, and second, to ensure the input current drawn from the AC source faithfully follows a sinusoidal pattern, maintaining unity power factor.

Achieving these dual objectives necessitates a sophisticated control architecture, typically involving two nested control loops: an inner current loop and an outer voltage loop. The inner current loop is responsible for shaping the inductor current, making it track a reference signal that is proportional to the input voltage waveform. Simultaneously, the outer voltage loop manages the overall output voltage regulation, generating the reference for the inner current loop. Understanding the design principles for these loops, including the derivation of relevant transfer functions and the selection of controller parameters, is crucial for developing robust and efficient PFC systems. This lecture will systematically break down these elements, from power balance considerations to the nuances of modulator operation and the critical criteria for controller design.

Core Concepts

Term	Definition	Significance
`Boost PFC Converter`	A power converter topology used to improve the power factor of an AC-DC power supply by shaping the input current.	Essential for meeting power quality standards and reducing harmonic distortion in the grid.
`Inner Current Loop`	The faster control loop responsible for regulating the	Ensures the input current is sinusoidal and in phase with

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	inductor current to follow a desired reference waveform.	the input voltage, achieving power factor correction.
`Outer Voltage Loop`	The slower control loop that regulates the converter's output DC voltage to a set reference value.	Provides stable DC output voltage, which is the primary function of the converter.
`Small Signal Modeling`	A linearization technique used to analyze the dynamic behavior of non-linear power converters around an operating point.	Allows for the derivation of transfer functions (like Gi(s) and Gv(s)) necessary for controller design using linear control theory.
`Modulator Block`	The component that translates the control signal (duty ratio) into a pulse-width modulated (PWM) signal for switching devices.	Converts the continuous control command into discrete switching actions, enabling power conversion.
`Carrier Signal`	A high-frequency, periodic waveform (e.g., triangular or ramp) used in PWM generation.	Provides the basis for comparison with the modulating signal (duty ratio) to determine switch on/off times.
`Bandwidth`	The frequency range over which a control system can effectively track a reference or reject disturbances.	A critical parameter for controller design, determining the speed of response and ripple rejection capabilities.
`Open Loop Transfer Function`	The product of the controller and plant transfer functions in a feedback system, before the loop is closed.	Its Bode plot (gain and phase) is used to analyze stability and performance characteristics of the closed-loop system.
`Phase Margin`	The amount of additional phase lag required to make a	A key indicator of system stability; a sufficient phase

Term	Definition	Significance
	system unstable at the gain crossover frequency.	margin (e.g., 35-60 degrees) ensures robust operation.
`Gain Margin`	The amount of gain increase required to make a system unstable at the phase crossover frequency.	Another stability indicator, ensuring the system remains stable even with variations in gain.

Detailed Analysis

1. Derivation of the Voltage Transfer Function (Gv(s))

The derivation of the voltage transfer function, Gv(s), which relates the output voltage (Vo(s)) to the inductor current (iL(s)), is crucial for the outer voltage loop design. This is achieved through a power balance approach, assuming that the input power equals the output power.

- **Power Balance Principle**: The instantaneous input power, derived from the AC input voltage (Vs,pk sinωt) and unity power factor current (Is,pk sinωt), is equated to the output power.
- Input Power: (Vs,pk / √2) * (Is,pk / √2) = (Vs,pk * Is,pk) / 2
- Output Power: Vo * iConverter (where iConverter is the average current flowing into the output stage).
- **Relating iConverter to iL**: In the positive half-cycle, the input current (is) is equal to the inductor current (iL), and in the negative half-cycle, is = -iL. Thus, the magnitude change of iL reflects the change in is.
- **Small Signal Analysis**: By assuming small variations around an operating point and setting the small signal variation of the input voltage (Vs,tilde) to zero, we can relate the small signal change in iConverter to the small signal change in iL.
- `iConverter(s) = (Vs,peak / (2 * Vo)) * iL(s)`
- **Output Stage Impedance**: The output stage consists of a parallel combination of the load resistor (RL) and the output capacitor (C). The impedance of this parallel combination is `(RL / (sC)) / (RL + 1/(sC)) = RL / (RLCs + 1)`.
- **Gv(s) Derivation**: Multiplying the iConverter(s) by the output stage impedance yields Vo(s).
- `Vo(s) = iConverter(s) * (RL / (RLCs + 1))`

- Substituting iConverter(s): `Vo(s) = (Vs,peak / (2 * Vo)) * iL(s) * (RL / (RLCs + 1))`
- Rearranging to find Gv(s) = Vo(s) / iL(s): `Gv(s) = (Vs,peak / (2 * Vo)) * (RL / (RLCs + 1))`
  This transfer function represents how changes in the inductor current affect the output voltage.

2. Understanding the Gain Block (Ki) and Modulator Operation

The Ki gain block connects the output of the current controller to the modulator, translating the control signal into a duty ratio.

- **Ki Definition**: Ki is defined as the ratio of the peak value of the control voltage (output of the current controller) to the peak value of the input current. It essentially scales the control signal to correspond to the duty ratio required to achieve a certain current.
- **Modulator Block Function**: The modulator block converts the continuous duty ratio (d) into a pulse-width modulated (PWM) signal that drives the converter switches (S1 and S2).
- **Carrier Signal**: A high-frequency carrier signal (e.g., triangular or ramp) is used. This carrier typically varies between 0 and a maximum control voltage (Vc).
- **Comparison**: The duty ratio (d), which varies between 0 and 1, is scaled to match the range of the carrier signal (0 to Vc). This scaled duty ratio (modulating signal) is then compared with the carrier signal.
- **PWM Generation**:
- If `Carrier > Modulating Signal`, then `PWM_S1 = 1` (switch S1 is ON).
- If `Carrier < Modulating Signal`, then `PWM_S1 = 0` (switch S1 is OFF).
- `PWM_S2` is typically generated by a NOT gate from `PWM_S1`.
- **Frequency Relationship**: The carrier frequency (fc) must be significantly higher than the frequency of the modulating signal (fm, which is the frequency of the duty ratio variation, typically 2fs or 100 Hz). This ensures that the PWM accurately reflects the desired duty cycle.

3. Two-Loop Control Structure and Controller Design Criteria

The closed-loop control of the PFC converter employs a nested two-loop structure: an inner current loop and an outer voltage loop.

- **Loop Hierarchy**:

- **Inner Current Loop**: This loop is designed to be much faster than the outer voltage loop. It tracks the instantaneous inductor current reference.
- **Outer Voltage Loop**: This loop is slower and regulates the DC output voltage. It generates the current reference for the inner loop. The output voltage is a DC quantity, and its changes are relatively slow, hence the slower response requirement.
- **Current Controller (Cc(s)) Design Criteria**:
- **High Gain at 2fs (Line Frequency Harmonics)**: The open-loop transfer function of the inner current loop (`Cc(s) * Ki * Gi(s)`) must have a very high gain at twice the line frequency (e.g., 100 Hz for a 50 Hz line frequency). This ensures that the actual inductor current accurately tracks the 2fs varying reference, which is crucial for unity power factor. A high gain at this frequency minimizes the error between the reference and the actual current.
- **Bandwidth Less Than Switching Frequency**: The bandwidth of the current loop must be significantly less than the switching frequency (fsw), typically `fsw / 10`. This ensures that the controller does not amplify the high-frequency current ripple inherent in the converter's operation. While the ripple exists due to switching, the controller should attenuate it rather than react to it, focusing on tracking the average current.
- **Sufficient Phase and Gain Margins**: The current loop must exhibit sufficient phase margin (e.g., 35-60 degrees) and gain margin to ensure stability and robust operation against disturbances and parameter variations.
- **Voltage Controller (Cv(s)) Design Criteria**:
- **Very High DC Gain (at s=0)**: The open-loop transfer function of the outer voltage loop (`Cv(s) * Gccl(s) * Gv(s)`) must have a very high gain at DC (s=0). This ensures precise regulation of the DC output voltage, minimizing steady-state error.
- **Bandwidth (20-40 Hz)**: The bandwidth of the voltage loop should be relatively low, typically in the range of 20-40 Hz. This is because the output voltage is a DC quantity, and a very high bandwidth is not required and could lead to an oversized or overstressed controller. This range allows for optimal sizing and performance.

4. Other Control Methods

While average current control with constant switching frequency is commonly used and discussed, other methods exist:

- **Peak Current Control**: In this method, the inductor current is allowed to rise until it hits a peak reference value, at which point the switch is turned off. The turn-off time might be constant, leading to a variable switching frequency, especially as the input voltage varies. This method ensures the peak current follows the reference.
- **Variable Hysteresis Control**: This technique defines a hysteresis band around the current reference. The switch state is toggled when the actual current crosses the upper or lower band limits. The band itself can be varied over the line cycle. This results in a variable switching frequency but keeps the current within a defined envelope.

Key Takeaways

- [ ] The Boost PFC converter's closed-loop control aims to regulate output voltage and achieve unity power factor by shaping input current.
- [ ] The control system uses a nested structure: a faster inner current loop and a slower outer voltage loop.
- [ ] The voltage transfer function Gv(s) is derived using power balance and small-signal analysis, relating output voltage to inductor current.
- [ ] The modulator block translates the duty ratio into PWM signals by comparing it with a high-frequency carrier signal.
- [ ] The inner current loop controller must provide high gain at twice the line frequency (e.g., 100 Hz) to track the current reference accurately.
- [ ] The current loop's bandwidth should be significantly lower than the switching frequency (e.g., fsw/10) to attenuate switching ripples.
- [ ] Both current and voltage loops require sufficient phase and gain margins for stable operation.
- [ ] The outer voltage loop controller needs a very high DC gain (at s=0) for precise output voltage regulation.
- [ ] The voltage loop's bandwidth should be relatively low (e.g., 20-40 Hz) to avoid over-stressing the controller.
- [ ] Alternative control methods like peak current control and variable hysteresis control offer different trade-offs in terms of switching frequency and ripple.

Conclusion

The detailed exploration of closed-loop control for single-phase Boost PFC converters highlights the critical interplay between theoretical modeling and practical design considerations. By meticulously deriving transfer functions like Gv(s) and understanding the role of the Ki gain block and the modulator, engineers can systematically approach the design of robust control systems. The emphasis on a two-loop structure, with a fast inner current loop and a slower outer voltage loop, is fundamental to achieving both power factor correction and stable output voltage regulation.

The specific criteria for designing the current and voltage controllers—such as ensuring high gain at 2fs for current tracking, attenuating switching frequency ripple, and maintaining adequate stability margins—are not merely academic exercises but practical necessities for reliable operation. These principles enable the creation of converters that not only deliver stable DC power but also draw current efficiently from the AC grid, minimizing harmonic distortion and improving overall power quality. As charging infrastructure continues to expand, the ability to design and implement such sophisticated control strategies will be paramount in ensuring the efficiency, reliability, and grid compatibility of future power systems.